

Abhishek Singh

Noida, Uttar Pradesh | [Abhi2001](#)

[+91-8126684451](#) | [abhi28031@gmail.com](#) | [Abhishek Singh](#)

WORK EXPERIENCE

Arcap REIT AI Solution / Data Scientist trainee

Aug 2025 – Jan 2026

- **Data Pipeline Development:** Developed automated data cleaning pipelines using Python and R, reducing manual data preparation time by 30%.
- **Feature Engineering:** Implementing advanced preprocessing techniques including categorical encoding, datetime extraction, and scaling/normalization to optimize data for machine learning models.
- **ML System Architecture:** Designing end-to-end cloud-based AI architecture using AWS (S3, SageMaker) and Azure for scenarios like Patient Risk Prediction.
- **Model Deployment:** Deployed Random Forest and CNN models via Flask, ensuring scalability and monitoring.

PROJECTS

Facial Mood Detection System / Python, Numpy, Matplotlib, OpenCV, TensorFlow & Keras

- Developed an end-to-end Face Emotion Detection system using a custom CNN (Conv2D, MaxPooling, Flatten, Dense layers) built and trained with TensorFlow/Keras, classifying facial expressions from grayscale preprocessed image data.
- Integrated OpenCV with the trained model to enable real-time emotion detection via webcam, applying the pipeline to use cases like smart surveillance, customer emotion analysis, and interactive AI applications.
- Achieved reliable multi-class emotion classification (happy, sad, angry, surprise, fear, disgust, neutral) by tuning CNN architecture and hyperparameters to improve validation accuracy and reduce overfitting.

Automatic Number Plate Recognition / Python, OpenCV, EasyOCR, Deep Learning

- Developed a real-time Object Detection for vehicle identification using YOLOV8 and EasyOCR, achieved 92% character recognition.
- Leveraged CNN-based architectures to reach 96% precision in license plate localization and segmentation.
- Optimized the inference pipeline to process high-definition video feeds at 30 FPS, ensuring suitable for real-time security applications.

Healthcare Chatbot for Patient Support / Python, AWS Lex, Lambda, DynamoDB, NLP

- Developed a full end-to-end healthcare chatbot designed to provide 24/7 patient support and automate routine medical inquiries.
- Leveraged AWS Lex to implement Natural Language Understanding (NLU), achieving high-precision intent classification for patient queries.
- Integrated conversation AI with automated workflows to streamline patient support tasks as per specific healthcare instruction sets.

TECHNICAL SKILLS

Languages: Python, R, SQL, C++, JavaScript

Computer Vision: OpenCV, YOLO, Image Segmentation, Image Processing

Deep Learning: TensorFlow, PyTorch, CNNs, Data Augmentation

Deployment & Cloud: Rest APIs, Flask, AWS(Sagemaker, S3, Lambda), Azure

Engineering: Data Structure & Algorithms, Git, Linux

EDUCATION

IIT Guwahati & Daksh Gurukul

Mar 2025 – May 2026

Credit-Linked Program in Data Science

JSS Academy of Technical Education(AKTU)

2020 – 2024

B.Tech - Computer Science and Engineering

Noida, India

ACHIEVEMENTS

- Certificate of Data Analytics Job Simulation with Deloitte.
- Completion Certificate of Machine Learning Course with Infosys Springboard.